PRIVGUARD AI

The Independent Security Layer for AI Applications

Secure every prompt. Govern every tool. Protect every interaction.

- Decoupling safety and security policy enforcement from probabilistic LLMs.
- Local-first privacy enforcement with sensitive-data redaction before model transmission.
- Cryptographic task-scoped capability tokens for autonomous agent tools.
- Moon Sanctum Labs | Founder: Aniket Nandi (Kolkata, India) | Buildathon 2026

PRIVGUARD AI | PROBLEM STATEMENT

The Problem: The Fallacy of Model Self-Governance

Why Generative AI is Inherently Vulnerable in Production

You would never let a web user enforce database permissions. Why trust an LLM to govern itself?

- Data Ingress Leakage: Customer PII, secrets, and proprietary code sent raw to cloud APIs.
- Adversarial Hijacking: Prompt injections and jailbreaks effortlessly override soft system prompts.
- Ungoverned Agent Tools: Autonomous agents invoke sensitive APIs without cryptographic boundaries.
- Compliance Vacuum: Storing prompts violates DPDP/GDPR; omitting logs blinds forensic investigations.

PRIVGUARD AI | PROPOSED SOLUTION

The Solution: Deterministic Out-of-Band Gateway

Independent Enforcement Between Applications, Models & Tools

Fail-closed security kernel operating independently of model hallucinations.

- Ingress Defense: Scans secrets (entropy + patterns) & tokenizes PII (,).
- Adversarial Quarantine: Halts prompt injections with HTTP 403 before model context is loaded.
- Tool Authorization: Issues single-use HMAC-SHA256 capability tokens for agent tools.
- Content-Free Auditing: Records keyed HMAC-SHA256 fingerprints—zero raw prompt storage.

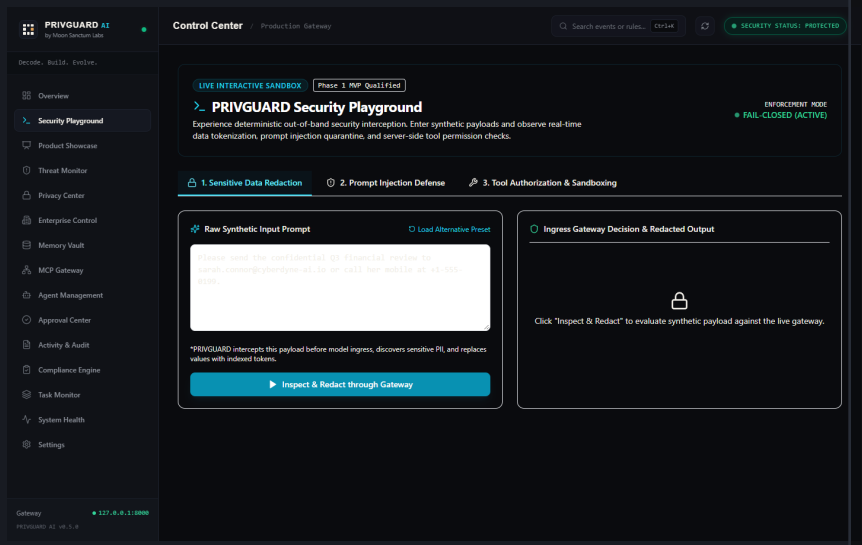
PRIVGUARD AI | DELIVERED PRODUCT

Current Product: Five Tested Phase 1 Capabilities

Completed MVP Verified Across 438+ Automated Backend Tests

Not a conceptual slide deck—a deployment-qualified security gateway.

- 1. Sensitive Data Protection: Presidio/regex tokenization & safe client-side de-tokenization.
- 2. Prompt Injection Defense: Deterministic signatures & heuristic quarantine (INJ_OVERRIDE_001).
- 3. Tool Authorization: Server-side ToolBroker with single-use task-scoped capability tokens.
- 4. Content-Free Activity Logs: Keyed HMAC fingerprints for DPDP/GDPR compliant forensics.
- 5. Policy Control Center: User-defined ALLOW/REDACT/BLOCK policies with real-time dry-run.



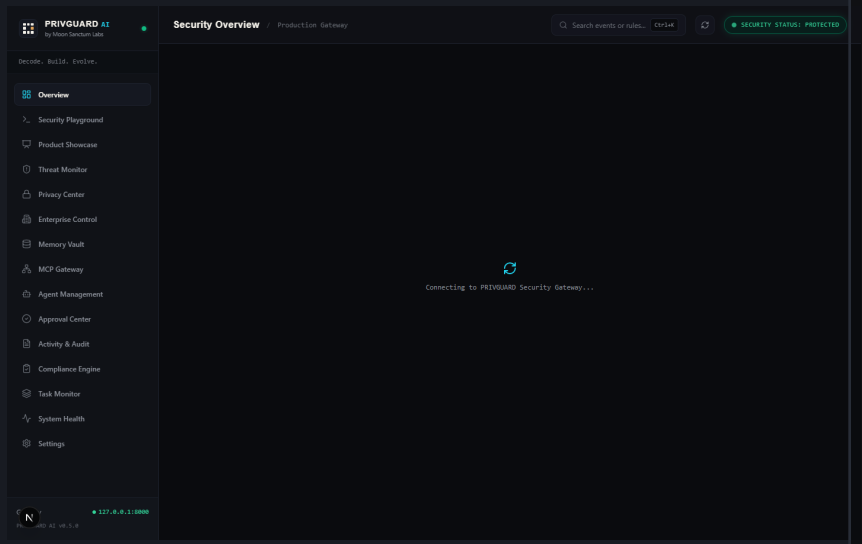
PRIVGUARD AI | PRODUCT SHOWCASE

Live Product: Executive Console & Security Playground

Real-Time Telemetry and Interactive Evaluation Sandbox

Full visibility into threats, redactions, and agent capabilities in real time.

- Interactive Playground: Evaluators test PII redaction, prompt injection, and tool checks.
- Executive Telemetry: Real-time counters for requests, blocked threats, and active agents.
- Live Local Integration: Tested end-to-end against local Ollama (llama3.2:3b).
- Zero Guesswork: Inspect exactly what was redacted and why a threat was quarantined.



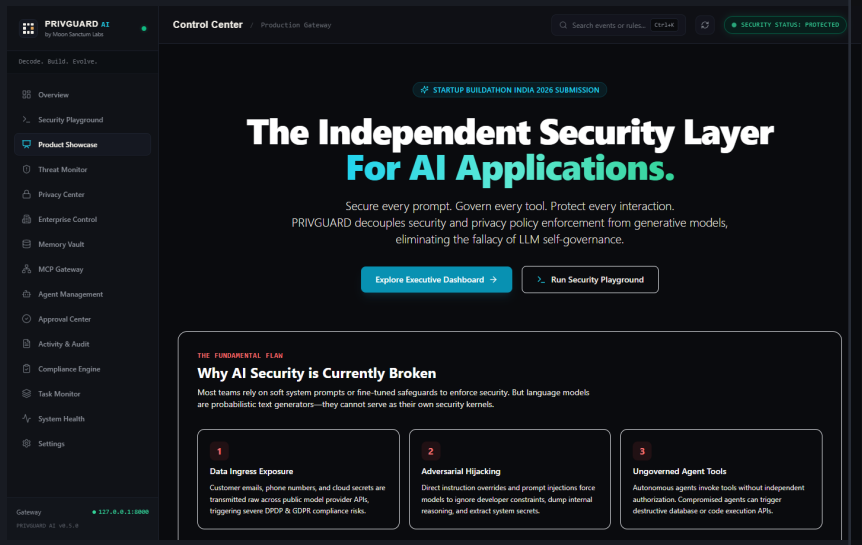
PRIVGUARD AI | TECHNICAL DESIGN

Technical Architecture & Enforcement Boundary

Application -> Ingress Pipeline -> ToolBroker -> Model -> Egress Inspection

Zero data egress in local mode; verified sensitive-data redaction in cloud mode.

- Client Layer: Standard OpenAI-compatible REST proxy interface (/v1/chat/completions).
- Ingress Pipeline: Secret scanning, prompt injection defense, PII tokenization.
- ToolBroker: Cryptographic HMAC capability validator with strict parameter constraints.
- Egress Pipeline: Model leakage inspection, placeholder restoration, content-free logging.



PRIVGUARD AI | COMPETITIVE ADVANTAGE

Market Differentiation: Why PRIVGUARD Wins

Architectural Superiority Over Traditional WAFs and Model Wrappers

Deterministic, local-first protection built specifically for the generative AI era.

- Enforcement Boundary: Operates at the AI application layer, not just generic network packets.
- Fail-Closed Philosophy: Malicious inputs and live secrets are blocked before reaching model APIs.
- Cryptographic Tool Grants: First gateway to treat agent tools as capability-gated primitives.
- Privacy by Design: Never stores raw user prompts, preventing secondary database leaks.

Market Opportunity & Business Model Hypotheses

Targeting the $15B+ AI TRiSM (Trust, Risk & Security) Market

Developer-first adoption scaling into enterprise compliance infrastructure.

- Beachhead ICP: AI SaaS startups and engineering teams deploying customer copilots.
- Regulatory Urgency: India DPDP Act 2023 (₹250 Cr penalty floor) & EU AI Act compliance.
- Community Tier (Free): Local-first gateway, SQLite storage, local Ollama, community rules.
- Team & Enterprise ($99 - $5,000/mo): Multi-tenant policies, SIEM export, air-gapped clusters.

Product Roadmap: The Three-Phase Vision

Clear Engineering Progression from Gateway to Mesh

What exists today, and what PRIVGUARD will become.

- Phase 1: Local-First AI Security Gateway (COMPLETED & QUALIFIED) - 438+ tests passed.
- Phase 2: Agent Security & Execution Mesh (PLANNED) - Multi-agent taint tracking & approval hooks.
- Phase 3: Privacy Infrastructure (LONG-TERM) - Confidential computing & zero-knowledge proofs.
- Commitment: Delivering verified, rock-solid engineering milestones before expanding.

PRIVGUARD AI | FOUNDER & EXECUTION

Founder & Execution Pedigree: Moon Sanctum Labs

Built from First Principles by Aniket Nandi (Kolkata, India)

Decode. Build. Evolve.

- Solo Founder: Full-stack systems engineer & product designer with obsessive focus on security.
- Verified Engineering: Single-handedly built gateway, test suite, and Next.js management console.
- Zero Security Vulnerabilities: Passed Bandit (0 high/medium) and Ruff clean static analysis.
- Vision: Establishing India's premier homegrown AI security infrastructure platform.